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Section 1.1 — Physical Quantities

SLO 1.1.1: Physical Quantities = Magnitude + Unit [E]

Core Definition

Every physical quantity in physics consists of exactly two inseparable parts that must always appear together:

Real-Life Analogy (Karachi Context)

Imagine you walk into a karyana store in Karachi and say *'I want 5 of sugar.'* The shopkeeper stares at you — 5 what? Kilograms? Grams? Teaspoons? The number 5 alone is completely meaningless. But '5 kg of sugar' is a complete, meaningful physical quantity.

Statement	Status	Reason
Speed is 60	INCOMPLETE	No unit specified
Speed is 60 km/h	COMPLETE	Both magnitude and unit given
Length is 2	INCOMPLETE	Is it 2 m, 2 cm, or 2 km?

■ ***Concept-Check Q: A student writes: 'The length of the table is 2.' What is wrong with this statement and why does it matter in physics?***

■ **Answer:** The unit is missing. Without a unit, we cannot determine whether the table is 2 metres, 2 feet, or 2 cm — completely different values. Both magnitude AND unit are always required.

SLO 1.1.2: Determining Appropriate Values of Physical Quantities [A]

What Does This Mean?

This SLO trains you to judge whether a calculated or given value is **reasonable** based on real-world experience — even before doing any mathematics. A good physicist always asks: *Does this answer make sense?*

Common Reference Values to Build Your Instinct

Physical Quantity	Reasonable Value	Unit
Height of a person	1.5 — 2.0	m
Mass of a student	50 — 80	kg
Room temperature	~300	K (≈ 27°C)
Speed of a car (city)	40 — 80	km/h
Time to fall 2 m	~0.6	s
Atmospheric pressure	~101,000	Pa

✔ **Exam Tip:** The unit can be correct but the magnitude still wrong! Always check BOTH together.

■ ***Concept-Check Q: A student calculates that a cricket ball falling from 2 m height takes 450 seconds to reach the ground. Is this value appropriate? Justify your answer.***

■ **Answer:** No — 450 seconds (7.5 minutes) is completely inappropriate. From everyday experience, an object dropped from 2 m hits the ground in approximately 0.6 seconds. The unit (seconds) is correct but the magnitude is absurd — the student made a calculation error.

SLO 1.1.3: SI Base Quantities and Their Units [R]

Why SI?

SI stands for *Système International d'Unités* (French for International System of Units). It is the globally agreed standard for measurement in science, ensuring that scientists worldwide speak the same measurement language.

The 7 SI Base Quantities

Base Quantity	SI Unit	Symbol	Nature
Length	metre	m	Base — independent
Mass	kilogram	kg	Base — independent
Time	second	s	Base — independent
Electric Current	ampere	A	Base — independent
Temperature	kelvin	K	Base — independent
Amount of Substance	mole	mol	Base — independent
Luminous Intensity	candela	cd	Base — independent

Memory Aid

Mnemonic: "Time — Lazy Men Travel Early, Keeping Many Clocks" → Time, Length, Mass, Temperature, Electric Current, Amount of Substance, Luminous Intensity

■ ***Note:*** *Base quantities are INDEPENDENT of each other and are the building blocks for ALL other (derived) quantities.*

■ ***Concept-Check Q: From this list — Mass, Speed, Temperature, Force, Electric Current — identify the SI base quantities and explain why the others are NOT base quantities.***

■ **Answer:** Base quantities: Mass, Temperature, Electric Current — they are independent and cannot be broken into simpler physical quantities. Speed = Length ÷ Time (derived). Force = Mass × Length ÷ Time^2 (derived). Both depend on other quantities, so they are NOT base quantities.

SLO 1.1.4: Derived Units as Products or Quotients of SI Base Units [E]

The Core Idea

Physics has hundreds of quantities, but only 7 base units. All other units are **derived** by multiplying or dividing base units. Think of base units as ingredients and derived units as the dishes you cook from them!

Step-by-Step Derivations

Quantity	Formula	Substitution	SI Unit
Speed	Distance ÷ Time	m ÷ s	ms^{-1}
Acceleration	Velocity ÷ Time	ms^{-1} ÷ s	ms^{-2}
Force (Newton)	Mass × Acceleration	kg × ms^{-2}	$kgms^{-2}$
Pressure (Pascal)	Force ÷ Area	$kgms^{-2}$ ÷ m^2	$kgm^{-1}s^{-2}$
Energy (Joule)	Force × Distance	$kgms^{-2}$ × m	kgm^2s^{-2}
Density	Mass ÷ Volume	kg ÷ m^3	kgm^{-3}

✔ **Exam Tip:** In AKU-EB exams, always derive the unit from the formula — never try to memorise. Start from the formula, substitute base units, simplify.

■ ***Concept-Check Q: Derive the SI unit of Density step by step. Given: Density = Mass ÷ Volume, Volume is in m^3, Mass is in kg.***

■ **Answer:** Density = Mass ÷ Volume = kg ÷ m^3 = kgm■^{3}. Note: Mass is on TOP (numerator) — not Volume!

SLO 1.1.5: Checking Homogeneity Using SI Base Units [A]

What is Homogeneity?

An equation is **homogeneous** (dimensionally consistent) when the units on the Left-Hand Side (LHS) are exactly equal to the units on the Right-Hand Side (RHS). Think of it as a balance scale — both sides must balance!

3-Step Method (Always Follow This)

Step	Action
1	Write the equation
2	Replace every quantity with its SI base units
3	Simplify both sides and check if they match

Worked Example: Check homogeneity of v = u + at

Term	Meaning	Units
v	final velocity	ms^{-1}
u	initial velocity	ms^{-1}
at	acceleration × time = ms^{-2} × s	ms^{-1}

Conclusion: LHS = ms■^{1} | RHS = ms■^{1} + ms■^{1} = ms■^{1} → LHS = RHS ✓ Equation is HOMOGENEOUS

■ Warning: Homogeneity checks ONLY confirm units are correct. They CANNOT verify numerical constants! Example: v = 2u + at is also homogeneous but physically WRONG. This is a favourite AKU-EB MCQ trap!

■ Concept-Check Q: Check homogeneity of: Pressure = Force ÷ Area (Force in kgms■^{2}, Area in m^2, Pressure in kgm■^{1}s■^{2})

■ **Answer:** LHS = kgm■^{1}s■^{2} | RHS = kgms■^{2} ÷ m^2 = kgm^(1-2)s■^{2} = kgm■^{1}s■^{2} → LHS = RHS ✓ Equation is HOMOGENEOUS

Section 1.2 — Dimensions

SLO 1.2.1: Concept of Dimensions [R]

What Are Dimensions?

Dimensions are one level more abstract than units. Instead of writing specific units like kg or metres, we write the **fundamental physical nature** of a quantity using capital letters. A dimension tells us WHAT TYPE of quantity something is, regardless of what unit system is used.

Dimension Symbols Table

Base Quantity	Unit	Dimension Symbol
Length	metre (m)	L
Mass	kilogram (kg)	M
Time	second (s)	T
Electric Current	ampere (A)	I
Temperature	kelvin (K)	θ (theta)

Units vs Dimensions — Key Difference

A Pakistani measures length in **metres**, an American in **feet**, an old British map uses **yards** — yet ALL represent the dimension **L**. The dimension never changes; only the unit changes!

Notation: Square brackets [ ] mean 'dimensions of'. Example: [Speed] = $LT^{■1}$ means the dimensions of speed are L divided by T.

■ Warning: CRITICAL: M = Mass only | L = Length only | T = Time only. NEVER write M for metres or S for seconds in dimensional analysis!

SLO 1.2.2: Dimensions of Physical Quantities [E]

Master Strategy

Always start from the formula → replace each quantity with its dimension symbol → simplify. Never try to memorise dimensions directly!

Complete Dimensions Reference Table

Physical Quantity	Formula	Dimensions
Length	—	L
Mass	—	M
Time	—	T
Area	L × L	L^2

Volume	L × L × L	L^3
Velocity	L ÷ T	LT^{-1}
Acceleration	LT^{-1} ÷ T	LT^{-2}
Force	M × LT^{-2}	MLT^{-2}
Work / Energy	MLT^{-2} × L	ML^2T^{-2}
Pressure	MLT^{-2} ÷ L^2	$ML^{-1}T^{-2}$
Momentum	M × LT^{-1}	MLT^{-1}
Power	ML^2T^{-2} ÷ T	ML^2T^{-3}
Density	M ÷ L^3	ML^{-3}

✔ **Exam Tip:** Work and Energy have IDENTICAL dimensions (ML^2T^{-2}). This is a popular MCQ question in AKU-EB — they represent the same physical concept!

■ ***Concept-Check Q: Find the dimensions of Momentum step by step. Given: Momentum = Mass × Velocity***

■ **Answer:** [Momentum] = M × LT^{-1} = MLT^{-1}

SLO 1.2.3: Homogeneity of Equations Using Dimensions [A]

Connection to SLO 1.1.5

This is exactly the same principle as SLO 1.1.5 — but instead of using SI units (kg, m, s), we now use dimension symbols (M, L, T). The method and golden rule are identical:

Worked Example 1: Check homogeneity of v = u + at

Term	Quantity	Dimensions	Simplified
v	velocity	LT^{-1}	LT^{-1}
u	velocity	LT^{-1}	LT^{-1}
at	LT^{-2} × T	$LT^{-2}\cdot T$	LT^{-1}

Result: [LHS] = LT^{-1} | [RHS] = LT^{-1} → Equation is HOMOGENEOUS ✓

Worked Example 2: Check homogeneity of KE = ½mv²

Term	Dimensions	Note
KE (energy)	ML^2T^{-2}	Given
m	M	mass
v^2	$(LT^{-1})^2 = L^2T^{-2}$	velocity squared

$\frac{1}{2}mv^2$	$M \times L^2T^{\blacksquare 2} = ML^2T^{\blacksquare 2}$	½ is dimensionless — ignored

Result: [LHS] = $ML^2T^{\blacksquare 2}$ | [RHS] = $ML^2T^{\blacksquare 2}$ → Equation is HOMOGENEOUS ✓

■ ***Note:*** *Pure numbers like ½, 2, π are DIMENSIONLESS — always ignore them during homogeneity checks.*

■ ***Concept-Check Q: Check the dimensional homogeneity of: s = ut + ½at² (s=displacement, u=velocity, t=time, a=acceleration)***

■ **Answer:** [s] = L | [ut] = $LT^{\blacksquare 1} \times T = L$ | [at^2] = $LT^{\blacksquare 2} \times T^2 = L$ → All three terms = L → Equation is HOMOGENEOUS ✓

SLO 1.2.4: Deriving Formulas Using Dimensional Analysis [A]

The Power of Dimensional Analysis!

This is one of the most powerful techniques in physics. If we know which physical quantities affect a quantity, we can derive the formula purely from dimensions — without knowing the physics! The only limitation is that we cannot find pure numerical constants like 2π.

6-Step Method (Memorise This!)

Step	Action
1	Identify all quantities involved
2	Write a general relationship with unknown powers (a, b, c...)
3	Write dimensions of both sides
4	Expand and collect powers of M, L, T
5	Compare powers — form simultaneous equations and solve
6	Substitute back to get the final formula

Worked Example: Time Period of a Simple Pendulum

The time period T depends on: length l, mass m, and acceleration due to gravity g [g] = LT■^{2}.

Step	Working
General relation	T = k · m■ · l■ · g■
Dimensions	T^1 = M■ · L■ · (LT■^{2})■
Expand RHS	T^1 = M■ · L^(b+c) · T^(-2c)
M: power 0=a	a = 0 (mass has NO effect!)
T: power 1=-2c	c = -½
L: power 0=b+c	b = ½
Final formula	$T = k\sqrt{(l/g)} \rightarrow T = 2\pi\sqrt{(l/g)}$

■ ***Note:*** *Dimensional analysis gives us $T = k\sqrt{(l/g)}$. Experiment tells us $k = 2\pi$. Dimensions cannot find the value of pure constants — this is their limitation.*

■ **Warning: Limitations: (1) Cannot find dimensionless constants like 2, π, ½. (2) Cannot tell if quantities are added or subtracted.**

■ ***Concept-Check Q: Derive the formula for the speed of sound v in a medium. It depends on Pressure P [ML■^{1}T■^{2}] and Density ρ [ML■^{3}]. Show all 6 steps.***

■ **Answer:** v = k·P■·ρ■ → LT■^{1} = (ML■^{1}T■^{2})■·(ML■^{3})■ → LT■^{1} = M^(a+b)·L^(-a-3b)·T^(-2a) From T: -2a=-1 → a=½ | From M: a+b=0 → b=-½ | From L: -½+3/2=1 ✓ Result: $v = k\sqrt{(P/\rho)}$

Section 1.3 — Precision and Accuracy

SLO 1.3.1: Differentiate Between Precision and Accuracy [R]

Feature	Accuracy	Precision
Definition	How close a measurement is to the TRUE/ACCEPTED value	How close REPEATED measurements are to EACH OTHER
Measures	Correctness	Consistency / Repeatability
Key Question	Is it correct?	Is it consistent?
Depends on	Systematic errors	Random errors
Example	Measuring 9.8 m/s^2 for g (true value = 9.8 m/s^2) — ACCURATE	Getting 9.2, 9.2, 9.2 m/s^2 every time — PRECISE but not accurate

The Dartboard Analogy

Imagine throwing darts at a dartboard where the bullseye is the TRUE value:

Scenario	Meaning
All darts near bullseye	Accurate AND Precise ✓✓
All darts clustered but away	Precise but NOT Accurate
Darts scattered near bullseye	Accurate but NOT Precise
Darts scattered everywhere	Neither Accurate nor Precise

■ ***Concept-Check Q: A student measures a table's length 5 times: 1.50m, 1.51m, 1.50m, 1.51m, 1.50m. The actual length is 1.80m. Is this data precise, accurate, both or neither?***

■ **Answer:** PRECISE but NOT ACCURATE. The readings are very consistent (clustered around 1.50–1.51m) showing high precision. However they are far from the true value of 1.80m, showing low accuracy.

SLO 1.3.2: Analysing Accuracy and Precision of Instruments [An]

How Instruments Affect Precision and Accuracy

Instrument	Least Count	Precision Level
Metre Rule	1 mm = 0.1 cm	Low precision
Vernier Callipers	0.1 mm = 0.01 cm	Medium precision
Screw Gauge	0.01 mm	High precision
Electronic Timer	0.001 s	Very high precision

■ ***Note:*** *A more precise instrument has a smaller least count — it can detect finer differences in measurement. Calibration ensures accuracy.*

Section 1.4 — Significant Figures and Uncertainty

SLO 1.4.1: Significant Figures and Uncertainty in Derived Quantities [A]

Rules for Counting Significant Figures

Rule	Example	Sig. Figs
All non-zero digits are significant	3456	4
Zeros between non-zero digits are significant	3406	4
Leading zeros (before first non-zero digit) are NOT significant	0.0045	2
Trailing zeros AFTER decimal point ARE significant	3.400	4
Trailing zeros WITHOUT decimal point are ambiguous	3400	2 or 4
Zeros after decimal in 0.00450	0.00450	3 (4, 5, 0 after decimal)

Uncertainty in Derived Quantities

Operation	Rule for Uncertainty
Addition / Subtraction	Add the ABSOLUTE uncertainties of each quantity
Multiplication / Division	Add the PERCENTAGE (fractional) uncertainties of each quantity
Power (x■)	Multiply percentage uncertainty by the power n

✔ **Exam Tip:** Your final answer should have the same number of significant figures as the LEAST precise measurement used in the calculation.

SLO 1.4.2: All Measurements Contain Uncertainty [E]

Why Is Uncertainty Inevitable?

- **1. No measuring instrument is infinitely precise** — every instrument has a least count which sets a lower limit on precision.
- **2. No physical system can be perfectly controlled** — temperature, vibrations, air currents all introduce small variations.
- **3. Human limitations** — reaction time, parallax error, and reading errors all contribute to uncertainty.
- **4. Fundamental quantum limit** — at the atomic scale, Heisenberg's Uncertainty Principle sets an absolute limit on measurement precision.

Types of Errors

Type	Definition	Can It Be Eliminated?
Systematic Error	Consistent error in one direction — affects accuracy	Yes — by calibration

Random Error	Unpredictable variations — affects precision	Reduced by repeating measurements
Human Error	Mistakes by the observer (parallax, reaction time)	Reduced by careful technique

■ ***Concept-Check Q: In one sentence, why do ALL measurements in physics contain uncertainty?***

■ **Answer:** All measurements contain uncertainty because no measuring instrument is infinitely precise and no physical system can be perfectly controlled, making some degree of error unavoidable.

Quick Revision Summary — Chapter 1

SLO	Topic	Key Point
1.1.1	**Physical Quantity**	Always = Magnitude + Unit. A number without a unit is meaningless.
1.1.2	**Appropriate Values**	Judge if a value makes real-world sense. Both magnitude AND unit must be reasonable.
1.1.3	**SI Base Quantities**	7 base quantities: Length(m), Mass(kg), Time(s), Current(A), Temp(K), Amount(mol), Luminosity(cd).
1.1.4	**Derived Units**	Formed by multiplying/dividing base units. Always derive from formula — never memorise.
1.1.5	**Homogeneity (Units)**	Units LHS = Units RHS. Cannot verify numerical constants — major MCQ trap!
1.2.1	**Dimensions**	M=Mass, L=Length, T=Time. Abstract representation independent of unit system.
1.2.2	**Dimensions Table**	Key: Force=MLT^{-2}, Energy=ML^2T^{-2}, Pressure=$ML^{-1}T^{-2}$, Momentum=MLT^{-1}
1.2.3	**Homogeneity (Dims)**	[LHS]=[RHS]. Ignore pure numbers (½, π). $s=ut+\frac{1}{2}at^2$ → all terms = L ✓
1.2.4	**Deriving Formulas**	6-step method using powers a,b,c. Cannot find pure constants. $T=k\sqrt{(l/g)}$ example.
1.3.1	**Precision vs Accuracy**	Accuracy=closeness to true value. Precision=consistency of repeated readings.
1.3.2	**Instruments**	Smaller least count = higher precision. Calibration ensures accuracy.
1.4.1	**Significant Figures**	Leading zeros NOT sig. fig. Trailing zeros after decimal ARE sig. fig.
1.4.2	**Uncertainty**	All measurements have uncertainty — no instrument is perfect, no system perfectly controlled.

AKU-EB Exam Practice Questions — Chapter 1

Section A — MCQ Style (1 mark each)

1. Which of the following is a SI base unit?

A) Newton B) Pascal C) Kelvin D) Joule

Answer: C) Kelvin

2. The SI unit of pressure in base units is:

A) $kgm^{\blacksquare 1}s^{\blacksquare 2}$ B) $kgms^{\blacksquare 2}$ C) $kgm^2s^{\blacksquare 2}$ D) $kgm^{\blacksquare 2}s^{\blacksquare 1}$

Answer: A) $kgm^{\blacksquare 1}s^{\blacksquare 2}$

3. The dimensions of momentum are:

A) $MLT^{\blacksquare 2}$ B) $ML^2T^{\blacksquare 1}$ C) $MLT^{\blacksquare 1}$ D) $M^2LT^{\blacksquare 1}$

Answer: C) $MLT^{\blacksquare 1}$

4. An equation is checked for homogeneity and found correct. This means:

A) The equation is definitely correct B) Units on both sides match C) Numerical constants are verified D) Both A and C

Answer: B) Units on both sides match

5. Which measurement shows high precision but low accuracy?

A) 9.80, 9.81, 9.79 m/s^2 (true=9.8) B) 8.20, 8.21, 8.20 m/s^2 (true=9.8) C) 9.0, 9.8, 8.9 m/s^2 (true=9.8) D) 9.8, 8.2, 10.1 m/s^2 (true=9.8)

Answer: B) 8.20, 8.21, 8.20 — consistent but far from true value

Section B — CRQ Style (2–3 marks each)

CRQ 1 (2 marks) Derive the SI base units of Power. Show your working step by step.

■ **Answer:** Power = Work ÷ Time → [Work] = $ML^2T^{\blacksquare 2}$ → [Power] = $ML^2T^{\blacksquare 2}$ ÷ T = $ML^2T^{\blacksquare 3}$ → SI unit = $kgm^2s^{\blacksquare 3}$

CRQ 2 (3 marks) Check the dimensional homogeneity of the equation: $F = mv^2/r$ where F=force, m=mass, v=velocity, r=radius.

■ **Answer:** [F] = $MLT^{\blacksquare 2}$ | [mv^2/r] = $M\times(LT^{\blacksquare 1})^2\div L = M\times L^2T^{\blacksquare 2}\div L = MLT^{\blacksquare 2}$ → [LHS]=[RHS] → Equation is HOMOGENEOUS ✓

CRQ 3 (2 marks) Differentiate between systematic and random errors with one example each.

■ **Answer:** Systematic error: consistent error in one direction, e.g. a wrongly calibrated balance always reads 0.5g too high — affects accuracy. Random error: unpredictable variation, e.g. slight variations in reaction time when using a stopwatch — affects precision.